

Smart Transit Navigation System

A centralized digital platform connecting passengers, drivers, transport authorities and for real-time public transportation management.

PROJECT REPORT · GROUP NO. 07

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Date of Submission: 20.08.2026

Why a Smart Transit Navigation System?

The Problem Today

- Passengers don't know a bus's current location or arrival time
- Route confusion and delayed service information
- Slow communication during emergencies
- Authorities struggle to monitor buses, fitness, crowd density and maintenance efficiently

Our Approach

- A centralized digital platform connecting three stakeholder roles on one system
- Passengers: search routes, GPS-track buses, view crowd density, women-only bus info
- Drivers: share GPS location, update status, report emergencies
- Authorities & owners: manage fleet, fitness, maintenance and operations
- Built on GPS tracking, a centralized database, role-based access and notifications

Existing Systems & the Gap We Address

1

GPS-Based Bus Tracking

Live vehicle location captured and transmitted centrally — realized here through the `GPSTLocation` component linked to the `Bus` class.

2

Passenger Information Systems

Typical systems cover routes, schedules and locations; ours extends this with crowd density, women-only bus info and emergency alerts.

3

Transport Management Systems

Admin platforms usually offer vehicle/route monitoring; ours adds `BusOwner` and `TransportAuthority` roles for fitness and maintenance reporting.

4

Emergency Communication

Drivers can report emergencies directly, while `TransportAuthority` and `TrafficPolice` monitor or receive the alerts in real time.

3. PROBLEM STATEMENT

An Integrated System Is Needed To..

- Provide real-time bus location information
- Help passengers find suitable routes
- Provide estimated arrival information
- Display passenger / crowd density
- Provide women-only bus information

- Allow drivers to report emergencies
- Monitor vehicle fitness and maintenance
- Send service and emergency notifications
- Centralize transport information for authorities
- Improve communication between passengers, drivers and authorities

4. OBJECTIVES

Main Objective & Specific Goals

Main Objective: To develop a Smart Transit Navigation System that improves public transportation information, monitoring, communication, and emergency response through a centralized digital platform.

1

Provide updated bus route information

2

Provide live bus location through GPS

3

Provide estimated arrival time

4

Show crowd density to passengers

5

Share women-only bus information

6

Let drivers update location & status

7

Enable emergency reporting by drivers

8

Support vehicle management for owners

9

Monitor fitness & maintenance records

10

Let authorities monitor operations

11

Deliver emergency & service notices

12

Maintain centralized transport records

5. PROPOSED SYSTEM

Three Roles, One Centralized Platform

Passenger

Search routes, track buses,
view crowd density, receive
notifications, women-only
bus info

Driver

Share GPS location, update
vehicle status, report
emergencies, receive route
guidance

Transport Authority

Manage users, monitor
operations, view transport
reports

- User Authentication assigns each user a role at login: Passenger, Driver, Transport Authority— driving what they can see and do.

Functional & Non-Functional Requirements

Functional (by role)

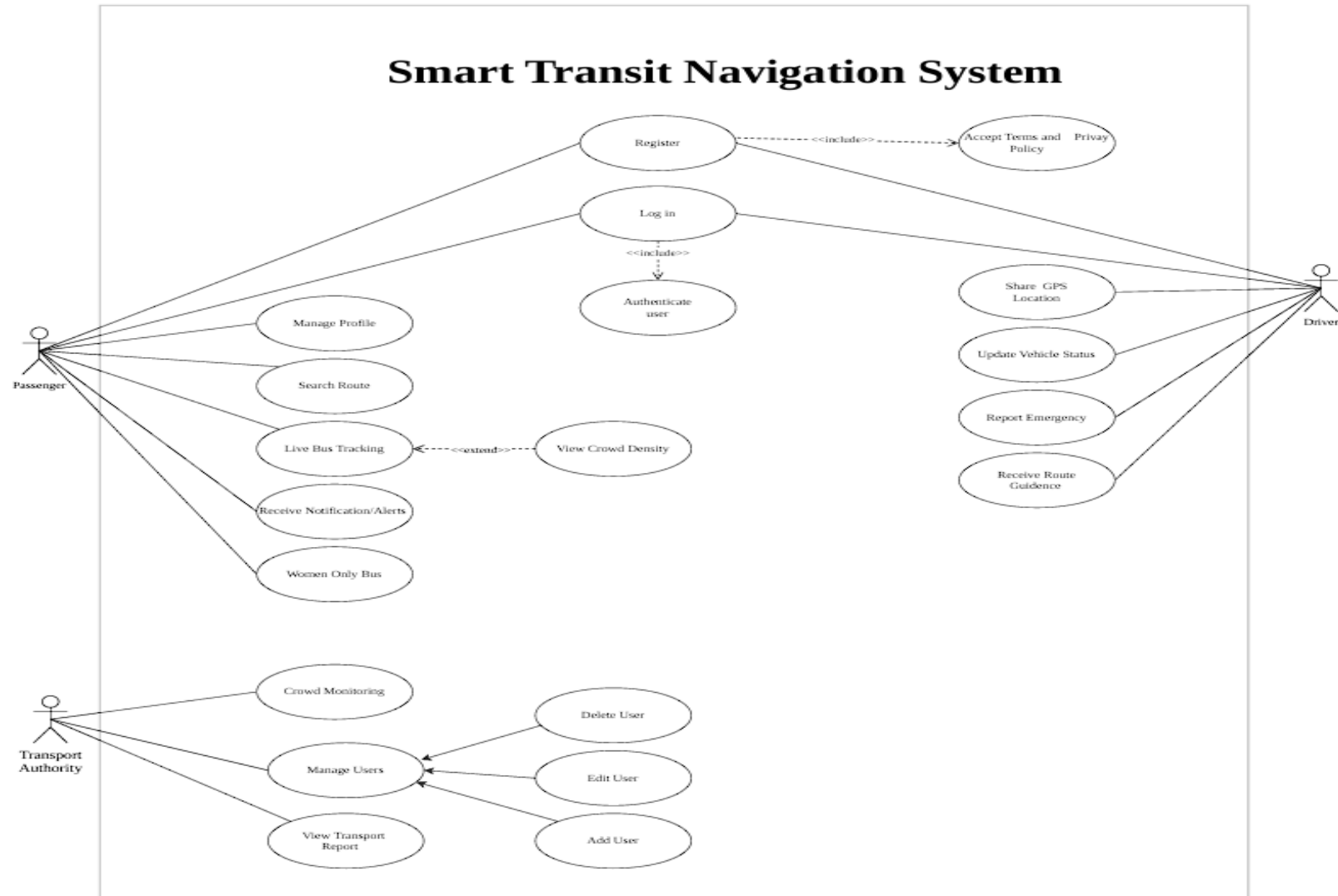
- Passenger: register/login, search routes, track buses, view crowd density & women-only info
- Driver: login, share GPS, update status, report emergencies, get route guidance
- Transport Authority: manage users, monitor operations, view reports
- System: store user/bus/route/GPS data, maintain emergency & maintenance records, generate notifications

Non-Functional

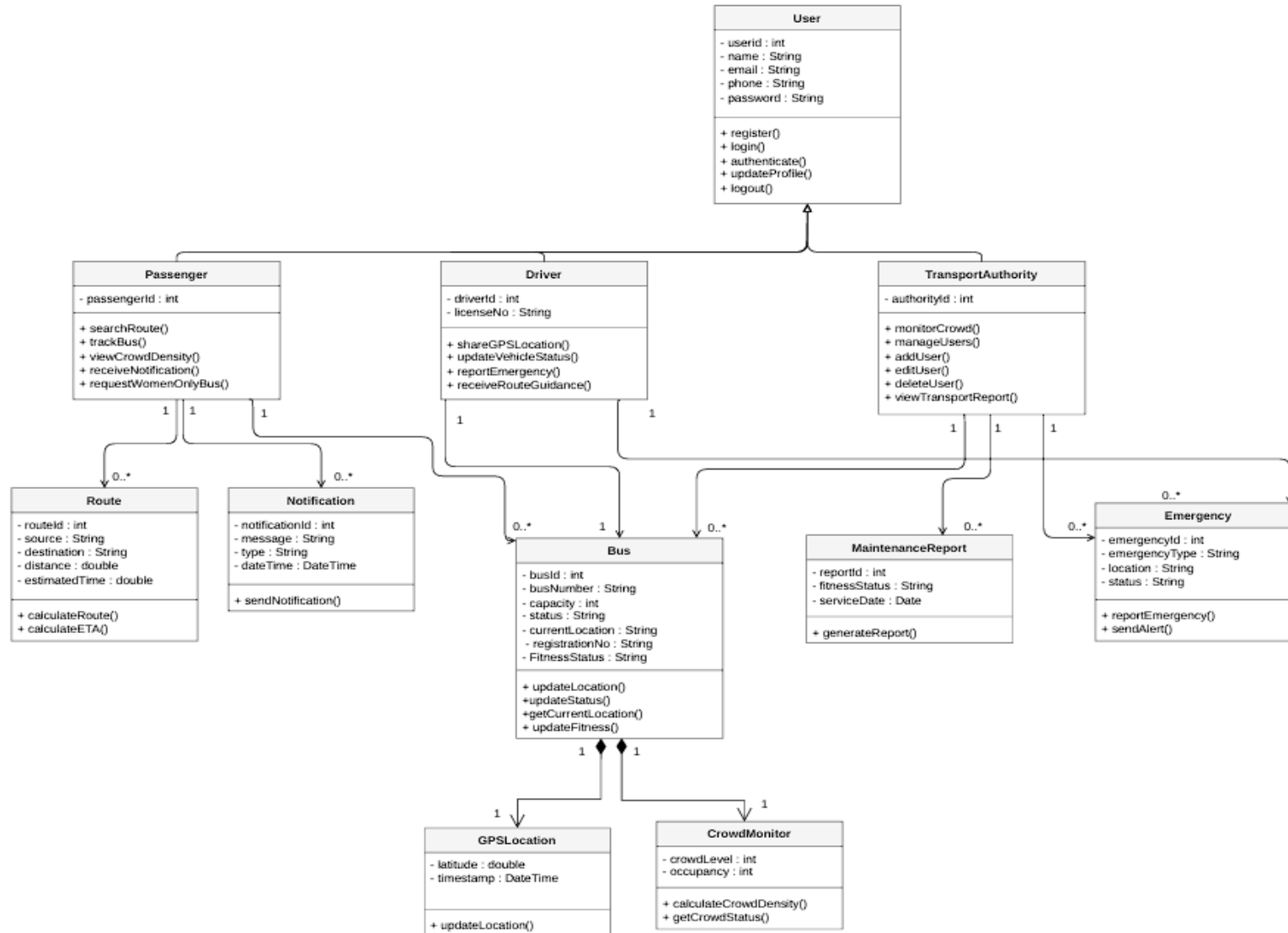
- Performance — minimal delay for location & service info
- Security — secure authentication, role-based access
- Reliability — reliable records of buses, routes, emergencies
- Availability — always accessible when information is needed
- Usability — simple interface for all user types
- Maintainability — supports future updates
- Scalability — supports growth in buses, routes and users

7.1 USE CASE DIAGRAM

Actors & Interactions



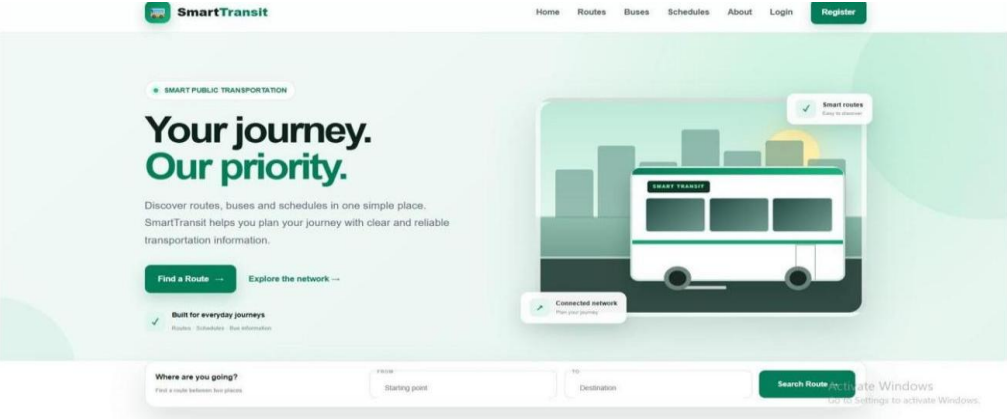
Core Entities & Relationships



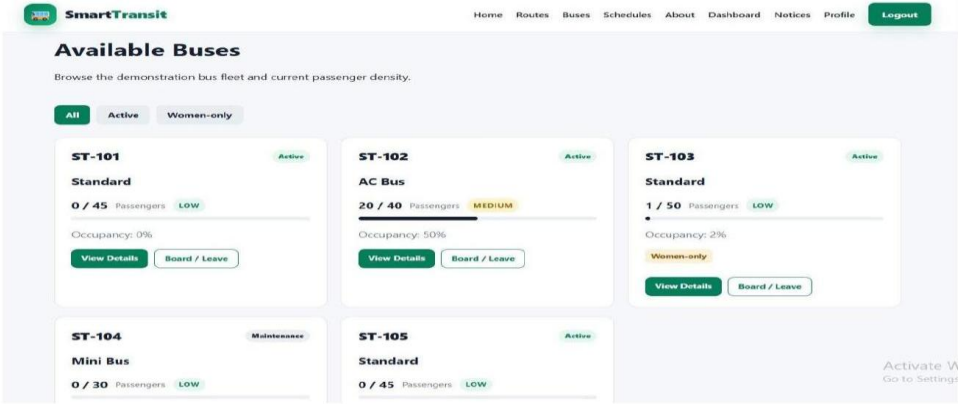
9. USER INTERFACE

The System in Action

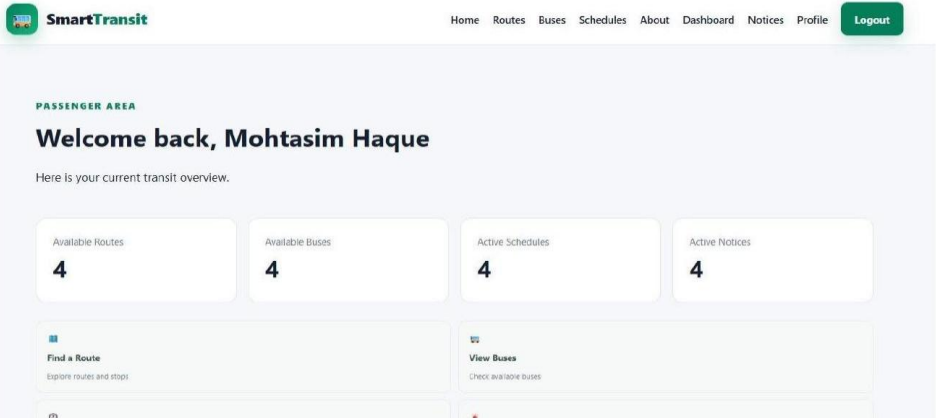
Home Page



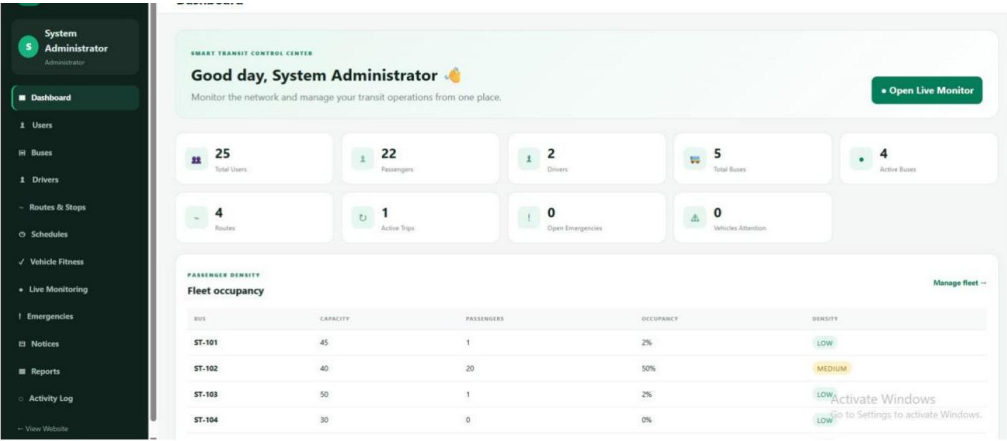
Available Buses & Boarding



Passenger Dashboard



Admin Control Center



What's Next for the Platform

Current Limitations

- Depends on stable internet for real-time GPS tracking
- GPS accuracy may drop in poor-coverage areas
- Relies on drivers regularly updating vehicle info
- Crowd density data may not always be fully accurate
- Currently covers buses only, not other transit modes
- Integration with existing infrastructure needs added resources
- Handling user & location data raises privacy considerations

Future Improvements

- AI-based traffic prediction for more accurate delay estimates
- Online ticket booking and digital payments
- Expand to trains, metro, and other public transport
- Multilingual support for wider accessibility
- IoT sensors for richer, real-time crowd analysis
- Dedicated Android & iOS mobile applications
- Route recommendations for fastest, least-crowded trips

13. CONCLUSION

A Practical Foundation for Modern Public Transit

The Smart Transit Navigation System brings passengers, drivers, bus owners, transport authorities, and traffic police onto one centralized platform — integrating real-time GPS tracking, route management, crowd monitoring, emergency reporting, vehicle fitness monitoring, and notifications.

It reduces uncertainty for passengers, improves stakeholder communication, and helps authorities monitor services more efficiently — a strong, practical base to build on despite some remaining limitations.

3

Stakeholder Roles

15

Planned Activities

65

Day Project Timeline



Thank You

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